

Al-Amin Hossain Nahid

Full-Stack Developer | CS Undergraduate

Ukil-Bari Road, Sreemangal | +8801305810184 | robin.nahid123@gmail.com

LinkedIn: al-aminnahid | **GitHub:** AlAminNahid | **Portfolio:** AlAmin

Summary

Computer Science undergraduate at the American International University–Bangladesh (AIUB) with hands-on full-stack engineering experience across web and Android development. Comfortable working end to end — designing relational data models, building and securing REST APIs, and developing the interfaces that consume them. Research interests in Machine Learning, Natural Language Processing, and Large Language Models.

Skills

- **Programming Languages:** C/C++, Java, C#, Python, JavaScript, TypeScript, Kotlin
- **Frontend:** HTML, CSS, Tailwind CSS, React, React Native, Next.js
- **Backend:** NestJS, FastAPI, ASP.NET, PHP
- **Databases & ORMs:** PostgreSQL, MySQL, MongoDB, TypeORM, Prisma, SQLAlchemy
- **Mobile:** Android SDK, Retrofit, MVVM, Material Design
- **AI/LLM:** REST-based LLM integration
- **Tools:** Git, Docker, Postman, Linux, Android Studio, Firebase, Vercel

Experience

AuraDev LTD | Software Engineering Intern

June 2026 - Present

- Contributing to **GrantOS**, an AI-assisted grant management platform, working end to end across a Next.js front end and a FastAPI backend.
- Developing REST API endpoints and business logic, with relational data models managed through SQLAlchemy and Alembic migrations on PostgreSQL.
- Building client-facing features including document workflows, rich-text editing, and authenticated multi-user access via Clerk.
- Integrating LLM APIs (Anthropic Claude, Google Gemini) into product features for proposal drafting and document analysis.
- **Technologies:** TypeScript, React, Next.js, FastAPI, Python, PostgreSQL, SQLAlchemy, Clerk, Anthropic Claude API

Projects

MessMasterApp_Backend | Backend Developer

GitHub

- Developed a RESTful backend for a smart mess management system using NestJS and PostgreSQL, deployed as a live API service.
- Designed role-based authentication and authorization with JWT (Access & Refresh Tokens) via Passport, bcrypt password hashing, and route-level guards separating manager and member permissions.
- Built modular APIs for mess management, meal tracking, utility expenses, notices, and member management using TypeORM entities and DTO validation with class-validator.
- Designed a normalized relational schema with versioned TypeORM migrations, supporting meal records, shared expenses, utility bills, and announcements.
- Added request rate limiting and transactional email flows for password reset and member onboarding.
- **Technologies:** TypeScript, NestJS, PostgreSQL, TypeORM, JWT, Passport.js, bcrypt, Nodemailer, Docker

MessMasterApp | Android Developer

GitHub

- Developed a native Android application that streamlines meal tracking, expense management, utility billing, and communication for shared living environments.
- Implemented role-specific interfaces for managers and members, consuming the backend REST API through Retrofit and OkHttp with token-based authentication.
- Structured the app on an MVVM architecture using ViewModel and Kotlin Coroutines for asynchronous data loading, with the Navigation Component managing multi-screen flows.
- Built responsive Material Design interfaces in Kotlin and XML — dashboards, meal management, utility tracking, notices, and profile management — using RecyclerView and ViewPager2.
- Automated meal-rate calculation and monthly bill generation through backend-synchronised records.

- **Technologies:** Kotlin, XML, Android SDK, MVVM, Retrofit, OkHttp, Coroutines, Navigation Component, Material Design

AI Chat | Full-Stack AI Web Developer

Live Demo | GitHub

- Developed a full-stack AI chat application with persistent multi-conversation history, backed by PostgreSQL through Prisma.
- Integrated the Google Gemini API behind server-side Next.js route handlers, keeping API credentials off the client.
- Designed REST endpoints for conversation create, read, update, and delete operations, with schema validation on incoming requests and typed error handling.
- Structured the application with a dedicated service layer, custom React hooks for chat and conversation state, and reusable UI components.
- **Technologies:** TypeScript, Next.js, React, Tailwind CSS, Prisma, PostgreSQL, Google Gemini API

Publications & Research

Accepted Publication

- **Al Amin Hossain Nahid**, Ferdus Hossain, Tariful Islam Fahim, and Salman Zzoha. “Intelligent Waste Classification for Sustainable Urban Development: A MobileNetV3-Based Deep Learning Approach Using the BDWaste Dataset.” To appear in *Proceedings of the International Conference on Power, Electronics, Communications, Computing, and Intelligent Infrastructure (PECCII 2026)*, 2026.

Ongoing Research

- **Al Amin Hossain Nahid**, Chowdhury Shakil Hasan, Rakibul Islam Akash, Abdullah Al Rafe Ayon. “An AI-Based Legal Advisory System for Traffic Violation Verification Using Official Traffic Regulations.” Manuscript in preparation.

Education

American International University-Bangladesh

B.Sc. in Computer Science and Engineering

Sept 2022 – Present

CGPA: 3.64/4.00

References

Tariful Islam Fahim

Data Engineer, STR search

Dhaka, Bangladesh

Phone: +8801534577441 | **LinkedIn:** tariful-islam